

LING487

Mathematics and Philosophy in Linguistics

Course syllabus — [Fall 2026]

Instructor:	Cem Bozsahin	Enrolment:	approximately 40
Meetings:	[Th 678]	Classroom:	[JF333]
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Format:	13 weeks; three contact hours per week; one midterm week		

Course description

This course asks what mathematics contributes to linguistic explanation, not merely how mathematical methods are applied after a linguistic analysis has been proposed. We shall study abstraction, invariants and variants, admissibility, functions and relations, formal grammars, automata, embedding, delineation, and the relation between models and linguistic reality. Throughout, formal techniques will be connected to empirical questions from morphology, phonology, syntax, semantics, language acquisition, and typology.

The course is designed principally for third- and fourth-year linguistics students. It assumes introductory courses in morphology and syntax, and some prior exposure to semantics. No mathematics beyond secondary-school or first-year university mathematics is presupposed. Willingness to work through notations and small formal problems is essential.

Learning outcomes

By the end of the course, students should be able to:

1. distinguish an abstraction from its instances, and an invariant from its variants;
2. use basic notation for sets, strings, relations, functions, composition, recursion, probability, and logic;
3. explain admissibility as a way of characterising linguistic structure;
4. construct and interpret small formal grammars, regular expressions, finite automata, push-down automata, and CKY charts;
5. compare a description, a grammar, a model, and a computation without treating them as interchangeable;
6. assess what a mathematical result does—and does not—establish about linguistic reality; and
7. relate formal results to empirical variation, acquisition, embedding, and the organisation of linguistic systems.

Required materials

- Cem Bozsahin, *Mathematics of Linguistic Reality: An Introduction* (hereafter *MOLR*).
- The accompanying exercise companion. Assigned exercises will be announced in class after the relevant material has been discussed.
- Occasional short supplementary readings, made available through the course site.

Students should read the assigned *MOLR* sections before class. The exercise companion is for practice, diagnosis, and discussion; completing a large number of exercises mechanically is not

the object of the course.

Weekly format

Most weeks combine two hours of lecture and guided analysis with a third hour devoted at the end to active work. In nine weeks, the third hour ends with one short, closed-device quiz designed to take no more than 20 minutes.

Assessment

Component	Weight
Nine in-class quizzes; best seven scores count (5% each)	35%
Midterm assessment	35%
Final assessment	30%
Total	100%

In-class quizzes

Each quiz asks one small question that can be answered within 20 minutes by a student who understands the week's central idea. A quiz may ask for a tiny construction, a counterexample, a comparison, or a brief interpretation; it will not reward the reproduction of long proofs or definitions. Quizzes are completed individually, in class, without notes, phones, computers, or other external assistance. Because only the best seven of nine scores count, two missed or unusually poor quizzes require no special remedy. Except where university regulations require otherwise, no make-up quizzes will be given.

It is natural to make mistakes in a quiz where you study what you have heard just an hour or two ago. This is how we learn. We are just checking progress to prepare for coming weeks.

Midterm and final assessment

The midterm checks command of the basic conceptual and formal vocabulary. The final assessment is not cumulative. Precise format and permitted materials will be announced in advance.

Use of generative AI and other assistance

The purpose of assessment is to establish what each student can reason through independently. Generative-AI systems, translation systems that generate or substantially rewrite answers, online solution services, and help from another person may not be used during quizzes, the midterm, or the final assessment.

Outside assessed work, students may use such tools for exploration and study, but remain responsible for checking every claim, symbol, derivation, and reference against the textbook and course discussion. A fluent answer is not evidence that an answer is sound. Whenever submitted coursework permits the use of such assistance, the instructions for that task will state

the required form of acknowledgement. Undisclosed generated work presented as one's own is subject to the university's academic-integrity rules.

Schedule

The schedule may be adjusted in response to the pace of the class. “Quiz” below means the short assessment held at the END of the third hour, 20 minutes maximum.

Wk	Focus	<i>MOLR</i> reading	Assessment
1	Abstraction and the mathematical basics	1.1–1.3: sets, strings, relations, functions, trees, composition, diagrams	—
2	Connectedness	1.4–1.7: Lambda calculus, recursion, probability, and logic	Quiz 1
3	Mathematics, computation, and linguistic structure	1.8–1.12: computation, linguistic structures, philosophy, typology, theories and models	Quiz 2
4	Linguistics as a special science	2.1–2.4: philosophy of science, ontogeny of grammaticality, fundamental questions, and mathematicisation	Quiz 3
5	Degrees of string admissibility	3.1–3.3: the Chomsky hierarchy; context-free, regular, and linear admissibility	Quiz 4
6	Describing admissible strings I	4.1–4.2: context-free and linear-indexed grammars	Quiz 5
7	Describing admissible strings II	4.3–4.4: regular expressions, linear grammars, and regular relations	Quiz 6
8	Consolidation and midterm	Review of Chapters 1–4.4	Midterm
9	Computing admissible strings I	4.5–4.6; 5.1–5.3: CKY, normal forms, pushdown automata and top-down/bottom-up parsing	Quiz 7
10	Computing admissible strings II	5.2, 5.4: finite automata, nondeterminism, conversions, and transducers	Quiz 8
11	Embedding and contingent homomorphisms	6.1–6.4: embedding, homomorphism, acquisition, diversity, morphology, and phonology	Quiz 9
12	Delineation, systems, and synthesis	6.5; return to the Preface roadmap and the course's central claims	—
13	Cumulative assessment and reflection	No new reading	Final assessment

Preparation, attendance, and classroom practice

The course depends on cumulative notation and discussion. Students are therefore expected to attend, bring the relevant reading and exercise work, and participate in reasoning through examples. Errors made openly while learning a formalism are useful evidence; unexplained answers are not. If illness, disability, religious observance, or another formally recognised circumstance affects participation or assessment, students should use the relevant university procedure and contact the instructor as early as practicable.

Academic integrity and accessibility

All work must comply with the university's current academic-integrity policy. Students requiring reasonable accommodations should contact the appropriate university office and the instructor. Course materials and assessment arrangements will be adjusted in accordance with approved accommodations.

This syllabus is a statement of the course design. Dates, institution-specific regulations, and assessment details announced through the official course site take precedence where a revision is necessary.